TrustSphere AI — Digital Evidence Verification Report

Generated 2026-10-23 16:43:03 UTC · Case ID 5c43fba8-aa68-4347-846a-eccf18df6f8d

1. Evidence Identification

Original filename	10_photo_matched_SPLICED.jpg
File type	image
SHA-256 hash	`4a1aecc15fd0a8ce47487bfcf8eb9744b345f87f35b9f8a9572056729d243854`
Uploaded (UTC)	2026-10-23T16:43:02.811200
Analysis status	done

2. Forensic Verdict

INCONCLUSIVE — overall authenticity score 0.49 (0 = fake, 1 = authentic)

Manipulation type	GAN / AI-Generated Image (confidence 0.76)
Generator attribution	GAN family (e.g. StyleGAN) (confidence 0.40)

Reasoning:
• Overall authenticity score 0.49 → Inconclusive.
• Deep classifier flagged synthetic imagery (p=0.76).

3. Agent Findings

metadata — authenticity 0.450

Check	Status	Note
camera_info	suspicious	No camera make/model — common in AI-generated or re-encoded images
software	ok	No editing software tag
timestamps	suspicious	No EXIF timestamps present

image_forensics — authenticity 0.520

Check	Status	Note
ela	ok	Mean error level 12.13 (heatmap saved)
fft	ok	High-frequency energy ratio 0.638
deep_model	suspicious	Deepfake classifier authenticity 0.243
xai_gradcam	ok	Model-attention heatmap saved (gradcam)

security — authenticity n/a (no signal)

Check	Status	Note
c2pa_manifest	ok	No C2PA content credentials embedded (typical for most media).
invisible_watermark	ok	No known AI-generator watermark detected.

4. Explainability Exhibits (XAI)

Exhibit — metadata feature importance (exact Shapley contributions; green → authentic, red → fake)

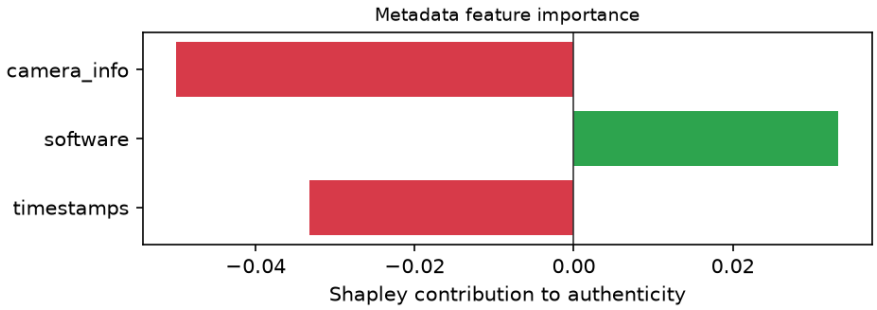
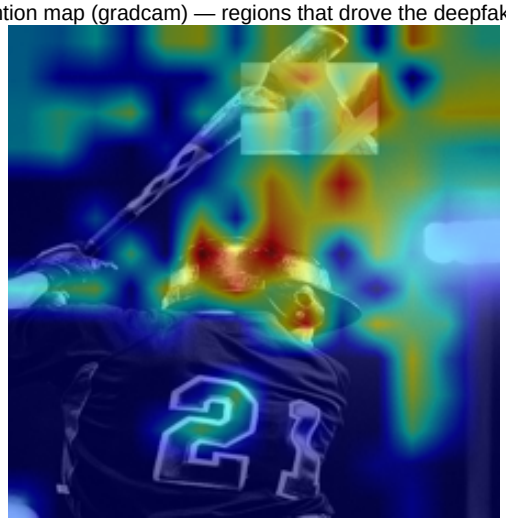


Exhibit — image_forensics: ELA heatmap (bright regions compress inconsistently → possible splice)



Exhibit — image_forensics: model attention map (gradcam) — regions that drove the deepfake classifier's decision



5. Chain of Custody

Event	Timestamp (UTC)	Detail
uploaded	2026-10-23T16:43:02.811200	
analyzed	2026-10-23T16:43:03.140156+00:00	
signed	2026-10-23T16:43:03.140297+00:00	`6217864d3e24af8f2157b4800b3b29c95f497c141297c36aae97212c7b77f184`
blockchain_skipped	2026-10-23T16:43:03.140312+00:00	`POLYGON_PRIVATE_KEY not set — anchoring disabled`

Blockchain network	polygon-amoy

Anchor status	skipped

6. Record Integrity — Digital Signature

The analysis record below is signed with the TrustSphere server's Ed25519 key. Any alteration of the signed payload invalidates the signature (verifiable via GET /api/verify/{file_id}).

Algorithm	Ed25519
Key fingerprint	`6217864d3e24af8f2157b4800b3b29c95f497c141297c36aae97212c7b77f184`
Signature	`7b52933cef8c410ff7da95818c876be89faf8c1208a99368063e8371adb2a47462e61ed567` `0468d8e54ce44606c866205e869010b33eaef6ec4878887b09f302`

This report was generated automatically by TrustSphere AI. The verdict is a statistical forensic assessment, not a legal determination.